CSCI 246 Midterm

October 7, 2026

Answer the questions below. For the counting questions (#3 and #6), you do not need to justify your answer.

1. Let x be an integer. Prove that $0 \mid x$ if and only if $x = 0$.
2. DeMorgan's laws are:

$$\neg(x \wedge y) = (\neg x) \vee (\neg y) \quad \text{and} \quad \neg(x \vee y) = (\neg x) \wedge (\neg y)$$

   Prove the first of these (using truth tables).
3. I want to create three playlists on my cell phone by downloading from a collection of 800 songs. One playlist is called "Exercise", another is called "Relaxing", and another is called "Studying". I want 10 different songs on each list. How many different ways can I load songs onto my phone if I allow a song to be on each playlist? And how many different ways can I load the songs if I want the three lists to have no overlap?
4. Let $A = \{n \in \mathbb{Z} : 3 \mid n\}$ and let $B = \{n \in \mathbb{Z} : 3 \mid (n + 6)\}$. Show that $A = B$.
5. Consider the relation $\leq$ on $\mathbb{Z}$. is this an equivalence relation? Justify your answer.
6. Ten people join hands for a circle dance. (a) In how many different ways can they do this? (b) Suppose five of these people are men, and the other five are women. In how many different ways can they join hands for a circle dance, assuming they alternate in gender around the circle?

Reference Material

- **Definition** (Divisible). Let a and b be integers. We say a is *divisible* by b, written $b \mid a$, provided there is an integer c such that $bc = a$.
- **Notation.** $\neg$ is logical not, $\wedge$ is logical and, $\vee$ is logical or.
- **Notation.** $\mathbb{Z}$ refers to the set of integers.